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ACADEMIC POSITIONS

2025–Present **Postdoctoral ERC Researcher, IFAE (Barcelona, Spain)**
ERC Synergy Grant GravNet, IFAE PI: Diego Blas
Establishing new experimental and theoretical approaches to fundamental physics using quantum sensors. Leading searches for new physics with magnetic levitation, including dark matter and exotic forces. Developing the theoretical and statistical framework for GravNet, a global network of electromagnetic cavities searching for high-frequency gravitational waves.

2022–2025 **Postdoctoral Associate, Rice University (Texas, USA)**
PI: Christopher D. Tunnell
Led the first search for ultralight dark matter using a magnetically levitated particle, worked on ultralight dark matter phenomenology with quantum sensors, and applied machine learning techniques to neutrino physics at direct detection experiments.

EDUCATION

2018–2022 **PhD** Astroparticle Physics
Institute for Particle Physics Phenomenology (IPPP)
Durham University, UK
Title: *"A New Era for Direct Detection Experiments: Probing New Neutrino Physics at Dark Matter Detectors with Solar Neutrinos"*
Supervisor: David G. Cerdeño
Awarded March 2023

2014–2018 **MPhys** Theoretical Physics (First Class Honours)
Durham University, UK
Dissertation Title: *"Novel Techniques for Dark Matter Direct Detection"*

PUBLICATIONS

† Supervised student in the work ★ Led the work

Preprints

1. *"Direct Detection of Solar Neutrinos to Probe Tensions and Break Degeneracies in Long-Baseline Neutrino Experiments"*, D. W. P. Amaral, D. Cerdeño, A. Cheek, V. Costa, P. B. Denton, and P. Foldenauer, arXiv:2610.18971 [hep-ph]. Submitted to PRD.

2. *"Multi-messenger signal from QCD-Axion Ultracompact Minihalos"*, D. W. P. Amaral, E. D. Schiappacasse, and K. Tapia-Rebolledo, arXiv:2608.25338 [hep-ph]. Submitted to PRD.

3. ★ *"First Search for Ultraheavy Dark Matter Using a Magnetically Levitated Particle"*, D. G. Uitenbroek, D. W. P. Amaral, J. Qin, J. Langendorff, A. Gingerich, T. H. Oosterkamp, and C. D. Tunnell, arXiv:2608.20464 [hep-ph]. Submitted to PRL.

4. ★ *"SNuDD: Solar Neutrinos for Direct Detection"*, D. W. P. Amaral, D. Cerdeño, A. Cheek, V. Costa, and P. Foldenauer, arXiv:2607.22817 [hep-ph]. Submitted to JHEP.

5. ★ *"A Levitated-Magnet Vector Force Sensor for Spin-Dependent Exotic Interactions"*, D. W. P. Amaral, L. Cong, T. M. Fuchs, H. Ulbricht, and D. Budker, arXiv:2606.30781 [hep-ph]. Submitted to PRL.

Peer-reviewed Publications

6. *"Global detector network to search for high-frequency gravitational waves (GravNet): conceptual design"*, D. Amaral et al., Phys. Rev. D, 10.1103/y1bf-4rfn (2026), arXiv:2603.24645 [astro-ph.IM].

7. *"Constraining dark photon dark matter with radio silence from soliton mergers around supermassive black holes"*, D. W. P. Amaral, E. D. Schiappacasse, and H.-Y. Zhang, JCAP **07**, 104 (2026), arXiv:2509.08932 [hep-ph].

8. † ★ *"Towards the direct detection of composite ultraheavy dark matter in quantum sensor arrays"*, D. W. P. Amaral, E. Cai, A. J. Long, J. Qin, and C. D. Tunnell, JCAP **06**, 035 (2026), arXiv:2512.10124 [hep-ph].

9. *"Model-independent searches of new physics in DARWIN with a semi-supervised deep learning pipeline"*, J. Aalbers et al., Eur. Phys. J. C **86**, 312 (2026), arXiv:2410.00755 [physics.ins-det], (DARWIN Collaboration, alphabetical author list including D. W. P. Amaral).

10. ★ *"Magnetic levitation as a new probe of non-Newtonian gravity"*, D. W. P. Amaral, T. M. Fuchs, H. Ulbricht, and C. D. Tunnell, Phys. Rev. D **113**, L021101 (2026), arXiv:2506.17385 [hep-ph].

11. *"Mechanical sensors for ultraheavy dark matter searches via long-range forces"*, J. Qin, D. W. P. Amaral et al., Phys. Rev. D **112**, 072003 (2025), arXiv:2503.11645 [hep-ph].

12. ★ *"First Search for Ultralight Dark Matter Using a Magnetically Levitated Particle"*, D. W. P. Amaral, D. G. Uitenbroek, T. H. Oosterkamp, and C. D. Tunnell, Phys. Rev. Lett. **134**, 251001 (2025), arXiv:2409.03814 [hep-ph].

13. ★ *"Fast Bayesian inference for neutrino non-standard interactions at dark matter direct detection experiments"*, D. W. P. Amaral, S. Liang, J. Qin, and C. Tunnell, Mach. Learn. Sci. Tech. **6**, 015049 (2025), arXiv:2405.14932 [cs.LG].

14. *"Rescuing the primordial black holes all-dark matter hypothesis from the fast radio bursts tension"*, D. W. P. Amaral and E. D. Schiappacasse, Phys. Rev. D **110**, 083532 (2024), arXiv:2312.09285 [hep-ph].

15. ★ *"Vector wave dark matter and terrestrial quantum sensors"*, D. W. P. Amaral, M. Jain, M. A. Amin, and C. Tunnell, JCAP **06**, 050 (2024), arXiv:2403.02381 [hep-ph].

16. *"Measuring the sterile neutrino mass in spallation source and direct detection experiments"*, D. Alonso-González, D. W. P. Amaral, A. Bariego-Quintana, D. Cerdeño, and M. de los Rios, JHEP **12**, 096 (2023), arXiv:2307.05176 [hep-ph].

17. *"A direct detection view of the neutrino NSI landscape"*, D. W. P. Amaral, D. Cerdeño, A. Cheek, and P. Foldenauer, JHEP **07**, 071 (2023), arXiv:2302.12846 [hep-ph].

18. *"Confirming $U(1)_{L_\mu - L_\tau}$ as a solution for $(g-2)_\mu$ with neutrinos"*, D. W. P. Amaral, D. G. Cerdeño, A. Cheek, and P. Foldenauer, Eur. Phys. J. C **81**, 861 (2021), arXiv:2104.03297 [hep-ph].

19. *"Solar neutrino probes of the muon anomalous magnetic moment in the gauged* $\mathrm{U}(1)_{L_\mu - L_\tau}$*"*, D. W. P. Amaral, D. G. Cerdeño, P. Foldenauer, and E. Reid, JHEP **12**, 155 (2020), arXiv:2006.11225 [hep-ph].

COLLOQUIA, CONFERENCE TALKS, AND SEMINARS

Colloquia

1. **University of Minnesota, Duluth**, *"Exploring Physics Beyond the Standard Model with Solar Neutrinos and Dark Matter Detectors"* (Aug. 2022).

Invited Seminars

2. **Quantum Seminar, JGU, Germany**, *"First Search for Ultraheavy Dark Matter Using a Magnetically Levitated Particle"* (Nov. 2026, scheduled).

3. **University of Victoria, Canada**, *"Searching for Dark Matter at the Extremes with a Levitated Magnet"* (Oct. 2026).

4. **CEICO Seminar, FZU, Czech Republic**, *"Floating into New Physics with Levitated Ferromagnets"* (May 2026).

5. **ICTP HECAP Seminar, Trieste, Italy**, *"Opening a New Window on Dark Matter with Magnetic Levitation"* (May 2026).

6. **Fermilab CPC Seminar, Batavia, USA**, *"Levitated Sensors at the Frontier of Fundamental Physics"* (Jan. 2026).
7. **KICP Group Meeting Talk, University of Chicago, USA**, *"Ultralight Vector Dark Matter"* (Jan. 2026).
8. **Leiden University, The Netherlands**, *"Building the Case for Magnetic Levitation in Fundamental Physics: Dark Matter and Non-Newtonian Gravity"* (Dec. 2025).
9. **Helmholtz Institute, Mainz, Germany**, *"Building the Case for Magnetic Levitation in Fundamental Physics: Dark Matter and Non-Newtonian Gravity"* (Nov. 2025).
10. **IFAE, Barcelona, Spain**, *"Magnetic Levitation for New Physics: From Dark Matter to Gravity"* (Nov. 2025).
11. **IPPP, Durham University, UK**, *"Hunting for Dark Matter Using Optomechanical Sensors"* (Jan. 2025).
12. **Rice University, Texas**, *"First Search for Ultralight Dark Matter Using a Magnetically Levitated Particle"* (Sept. 2024).
13. **Mitchell Institute, Texas A&M**, *"A Direct Detection View of the Neutrino NSI Landscape"*, video (Mar. 2023).
14. **Rice University, Texas**, *"Discovering New Neutrino Physics in Dark Matter Direct Detection Experiments with Solar Neutrinos"*, video (Mar. 2022).

Invited Conference and Workshop Talks

15. **Listening to the Cosmos: New Frontiers in Gravitational Wave Physics Workshop, GGI, Italy**, *"GravNet: Global detector network to search for high-frequency gravitational waves"* (Sept. 2026).
16. **Mitchell Conference 2025, Texas A&M**, *"Searching for Dark Matter and Fifth Forces Using Magnetic Levitation"* (May 2025).
17. **TACOS 2023, Rice University**, *"Surfing Dark Matter Waves at the Windchime Experiment"*, slides (Oct. 2023).
18. **Magnificent CEνNS, Munich**, *"A Direct Detection View of the Neutrino NSI Landscape"*, video (Mar. 2023).
19. **MultiDark 2022, Madrid**, *"Probing Neutrino Non-Standard Interactions in Dark Matter Direct Detection Experiments"*, slides (May 2022).
20. **UK HEP Forum 2021**, *"Catching a Criminal: Confirming $U(1)_{L_\mu - L_\tau}$ as the Solution to $(g-2)_\mu$ with Neutrinos"*, slides (Nov. 2021).
21. **WIN2021**, *"Exploring the $U(1)_{L_\mu - L_\tau}$ Solution to the Muon's Anomalous Magnetic Moment Using Future Experimental Probes"*, slides (July 2021).

Parallel Conference Talks

22. **PLANCK 2026, CERN, Switzerland**, *"Searching for Ultraheavy Dark Matter with Magnetic Levitation"*, slides (May 2026).
23. **TeVPA 2025, Valencia**, *"Floating into the Unknown: Levitated Sensor for New Physics"*, slides (Nov. 2025).
24. **Pheno 2025, Pittsburgh**, *"Magnetic Levitation for Fundamental Physics: From Dark Matter to Non-Newtonian Gravity"* (May 2025).
25. **TAUP 2023, Vienna**, *"Riding Dark Matter Waves at Windchime"*, slides (Aug. 2023).
26. **SUSY 2023, Southampton**, *"Towards the Detection of Ultralight Dark Photon Dark Matter Using Optomechanical Sensors"*, slides (July 2023).
27. **Pheno 2023, Pittsburgh**, *"Detecting Ultralight Dark Photon Dark Matter Using Optomechanical Sensors"*, slides (May 2023).
28. **Texas Section of the APS, Rice University**, *"Constraining Neutrino Non-Standard Interactions in Dark Matter Direct Detection Experiments"* (Oct. 2022).
29. **YTF 20**, *"Machine Learning Surrogates for Rapid Dark Matter Direct Detection Calculations"*, slides (Dec. 2020).
30. **BUSSTEPP 2019**, *"Using Machine Learning and AI Techniques for the Direct Detection of Dark Matter at SuperCDMS"*, PhD forum talk (Aug. 2019).

PRESS COVERAGE

"First search for ultraheavy dark matter with a magnetically levitated particle" (2026): Rice News (quoted), republished by Phys.org; IFAE News; Hackaday.

"First search for ultralight dark matter with a magnetically levitated particle" (2025): *Physics World* (interviewed); Rice News (quoted); Leiden University News; *The Debrief* (quoted).

SUPERVISING & TEACHING EXPERIENCE

Supervision

2026–Present **Martí Abella Cortes** (Master's Student, IFAE)
Supervising Master's thesis project on searching for ultralight axion-like dark matter with magnetic levitation experiments, with a paper in preparation. Thesis Title: *"Searching for Ultralight Axion-like Dark Matter with a Levitated Magnet"*.

2026 **Martí Abella Cortes & Fatih Žgalj** (Undergraduates, IFAE)
Co-supervised summer students on project exploring axion-matter effects.

2025–2026 **Ferran Serra Vancells** (Master's Student, IFAE)
Supervised Master's thesis project on detecting gravitational waves using magnetic levitation experiments.
Thesis Title: *"Detecting Gravitational Waves with Magnetic Levitation"*.

2024–Present **Yitang Chen** (Undergraduate, Rice University)
Supervising projects on magnetic levitation for new physics searches and ultralight dark matter. Currently working on a publication in the manuscript stage.

2022–2025 **Erqian Cai** (Undergraduate, Rice University)
Supervised a summer research project and co-supervised a Bachelor's thesis with Andrew Long on quantum sensing for dark matter detection, resulting in the publication of *"Towards the direct detection of composite ultraheavy dark matter in quantum sensor arrays"*, JCAP **06** (2026) 035.

2019–2021 **Tutor, Durham University**
Level 1 Foundations of Physics
Personal tutor for two groups of 6 students each year, leading tutorials and marking assignments and examinations.

Teaching

2025 **Statistics for Physicists Lecturer, Rice University**
Delivered part of a statistics for physicists lecture course, teaching hypothesis testing in a frequentist setting.

2024 **QFT Guest Lecturer, Rice University**
Delivered part of a Quantum Field Theory lecture course.

2018–2019 **Demonstrator, Durham University**
Level 2 Mathematical Methods in Physics
Demonstrated problem-solving techniques alongside course lecturer, marking assignments and examinations.

2018 **Assessment Developer, Durham University**
Level 2 Laboratory Skills and Electronics
IPPP Summer Internship: Developed course material as Jupyter notebooks
Supervisor: Dr D. Maître.

2017 **Course Materials Developer, Durham University**
Level 3 Foundations of Physics 3A
IPPP Summer Internship: Designed a series of interactive Jupyter notebooks demonstrating core nuclear and particle physics concepts.
Supervisor: Dr D. Maître.

INDUSTRIAL RESEARCH EXPERIENCE

2021 **Geoptic, Research & Development**
Developed machine learning tools for use in muography, including neural networks for classification tasks and variational autoencoders for anomaly detection.

2019 **IBEX Innovations, Research & Development**
Created a Python package for medical X-ray imaging for contrast enhancement, detail enhancement, and noise reduction. Results used in a successful clinical trial.

SCIENTIFIC SERVICE

Journal Reviewer: Reviewed for PRD and EPJC.

Workshop Organiser (February 2027): Organising the 5th Workshop on Ultra-high Frequency Gravitational Waves in Barcelona, Spain.

Seminar Series Organiser, IFAE (2025–2026): Organised weekly seminar series featuring both internal and external speakers.

Graduate Mentor, IFAE (2025–Present): Providing academic and pastoral support for a graduate student.

Member of EDI Committee, IFAE (2025–Present): Helping to improve EDI commitments at IFAE.

Journal Club Organiser, IFAE (2025–2026): Led weekly journal club.

Texas Section of the APS 2023: Session chair and judge.

TACOS 23: Member of organising committee for the TACOS 23 workshop. Session chair.

YTF 21: Chair of organising committee. Led the organisation of the YTF 21 conference.

YTF 20: Member of organising committee for the YTF 20 conference.

OUTREACH

Mathematics Symposium Invited Talk, Ramsey Grammar School (2025): Gave an invited colloquium-style talk on my research to A-level students (ages 17–18) attending an Island-wide mathematics competition.

Career Development Invited Talk, University of Houston (2023): Gave an invited colloquium-style talk on my research to undergraduate and graduate students. Spoke about pursuing the academic career path.

Career Development Invited Talk, Rice University (2022): Spoke to graduate students about pursuing the academic career path.

Biosciences Diversity Panelist, Rice University (2022): Spoke to undergraduate and graduate students about equality and diversity in the natural sciences.

Nuffield Research Placement Supervisor, Durham University (2022): Co-wrote a successful research proposal to host two Sixth-form students for a summer placement. Co-supervised a research project on neutrino physics beyond the Standard Model, leading to research talks in the UK.

Durham University Open Day Representative (2019): Spoke to prospective students about the physics course and undergraduate experience at Durham University.

Schools' Science Festival (2019): Volunteer at regional school event. Spoke to students about how direct detection experiments search for dark matter.

AWARDS & HONOURS

Three Minute Thesis Competition, Finalist (2022)
Finalist at public speaking competition for communicating thesis research.

Durham Excellence in Learning and Teaching Award (2021)
Led to a successful application to the Higher Education Academy for Associate Fellowship.

PROFESSIONAL AFFILIATIONS

2021–Present Associate Fellow of the Higher Education Academy (AFHEA)

Member of the **GravNet** collaboration.

SKILLS & SOFTWARE

Software: Developer of the open-source Python package `SNuDD` (Solar NeUtrinos for Direct Detection) to compute solar neutrino rates at direct detection experiments within and beyond the Standard Model.

Computing: Python, Mathematica, C, C++.

Statistics & Machine Learning: Bayesian inference, normalising flows, GPU acceleration and automatic differentiation, frequentist hypothesis testing.

Languages: English (native), Portuguese (native), Italian (competent), French (basic), Spanish (basic).